
Interpreting Imagined Futures: Discovering How World Action Models Steer Robot Behavior

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Abstract

World action models produce predicted futures alongside robot actions, but co-generation alone does not causally establish how the predicted future affects the action. Prior work shows that behavior depends on information at predicted-future token positions, but not how a particular future shapes the resulting action. We use two intervention stages to characterize that influence: Stage 1 measures how strongly and in what direction future content changes behavior, and Stage 2 identifies an internal pathway carrying that effect. Stage 1 inserts the future representation from one native rollout (the *donor*) into another from the same state (the *recipient*), holds the present and recipient action randomness fixed, and asks whether the recomputed action identifies the donor action, which the model never sees. In an outcome-unfiltered four-source evaluation across 90 Cosmos 3 states, every one of 1,080 off-diagonal transplants identifies the correct source. The same test succeeds in 1,439/1,440 FastWAM, 360/360 DreamZero, and 269/360 LingBot-VA comparisons, spanning joint and ordered video-first inference. Cosmos Policy further shows that this action shift survives one-chunk execution. In Stage 2, we independently cross the visible future and future-token attention K/V sources in Cosmos 3. Actions follow the K/V source in all 42 crossed comparisons; changing K/V source shifts donor projection by 0.885–0.895, while donor future with recipient K/V retains a smaller 0.119 residual. These interventions show that model-native future representations can exert source-specific control over predicted action and identify future-token K/V as a major, incomplete route in Cosmos 3.

1 Introduction

Many world action models (WAMs) produce predicted futures alongside robot actions [8, 17]. Joint generation, however, does not imply that action computation uses the predicted future: a policy could map the current observation and instruction directly to an action while producing a plausible future as an auxiliary output. Alternatively, predicted-future representations may serve as a computational workspace, encode a prospective robot pose for inverse dynamics [1, 7], or represent consequences used to evaluate actions [2, 3].

RIFT provides the closest prior causal evidence by masking reads from and editing values at predicted-future positions in several WAMs [23]. These interventions establish that behavior depends on information carried through the predicted-future interface. Because disrupting the interface can move an action in any direction, however, they do not reveal how a particular coherent future shapes behavior. We adapt mechanistic interventions to this joint video–action modality to make directional, testable claims about both future content and its computational pathway.

We ask three questions: Does a native future representation direct action; does this effect recur across inference organizations; and which internal pathway carries it? In Stage 1, we insert one rollout’s

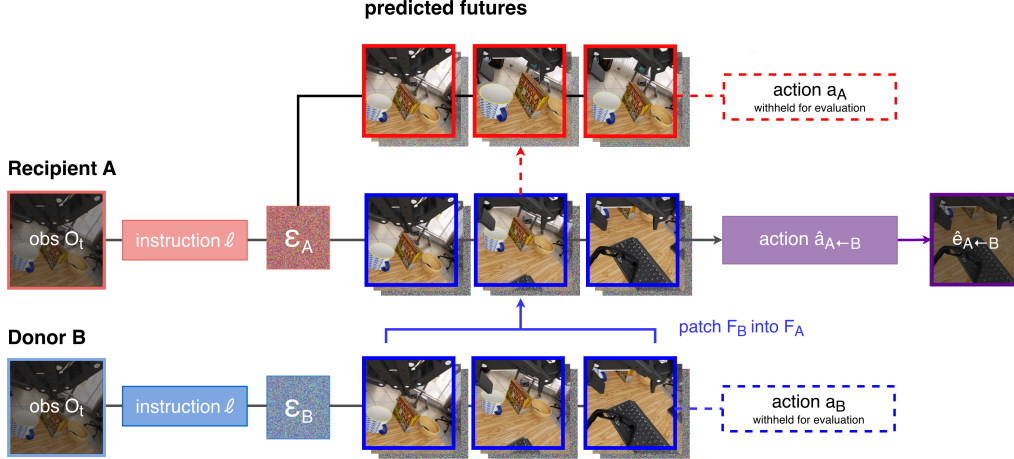


Figure 1: Future-content intervention. From native rollouts at the same state, Stage 1 inserts donor B 's future representation into recipient A while holding the present and recipient action randomness fixed. We test whether the recomputed action and, where executed, endpoint move toward B ; the donor action is withheld and used only for evaluation. The two-source schematic is repeated over complete four-source grids where supported.

37 future representation into another rollout from the same state. The receiving rollout is the *recipient*;
 38 the source is the *donor*. We recompute the recipient action while fixing the present and recipient
 39 action randomness. The donor action is withheld and used only as a directional reference. In Stage
 40 2, we cross the visible future and future-token attention K/V sources in Cosmos 3, separating what
 41 is shown from what action queries read.

42 The experiments yield three findings. First, an outcome-unfiltered four-source Cosmos 3 evalua-
 43 tion recovers the installed source from the recomputed action in all 1,080 off-diagonal comparisons
 44 across 90 states; FastWAM, DreamZero, and LingBot-VA replicate source-specific control across
 45 joint and ordered video-first inference. Second, Cosmos Policy's effect survives simulator execution
 46 and is concentrated in visual future targets, especially the wrist camera, whereas isolated robot or ob-
 47 ject pixels cannot reproduce Cosmos 3's complete-video effect. Third, Cosmos 3 actions follow the
 48 future-token K/V source in all 42 crossed comparisons. Together, these results show reproducible
 49 counterfactual control by native future representations and identify a major internal route for that
 50 control.

51 **Related work.** Visual robot policies may infer actions from predicted video, condition on its rep-
 52 resentation, or generate futures and controls jointly [1, 6–8, 10, 11, 17, 21]. These objectives do not
 53 establish how a particular future participates in inference. Recent work tests whether explicit rollout
 54 is needed [11, 22], aligns foresight and action representations [18], or intervenes on predicted-future
 55 positions [23]. Activation and path patching test whether a representation or route affects output
 56 [5, 13, 19]. We use within-state counterfactuals and directional source identification to determine
 57 what future representations control and to trace one route through predicted-future K/V [24].

58 2 Two-Stage Intervention Design

59 2.1 Stage 1: Future-content intervention

60 Starting from state S , native rollouts produce future representations F_j and paired actions $\mathbf{a}_j$ (Fig-
 61 ure 1). We recompute recipient i while replacing only its future representation with F_j . Observa-
 62 tion, instruction, recipient action randomness, sampler, and action coordinates remain fixed. Model-
 63 specific insertion follows the released inference route: step-matched future clamping for Cosmos
 64 3, matched donor latent-trajectory replay for DreamZero, and the future-derived cache for ordered
 65 video-first models. Cosmos Policy additionally encodes the endpoint of a native executed action,
 66 while Cosmos 3 tests predicted latents and executed videos. Appendix A.2 gives exact mechanics.

67 For action or endpoint $\mathbf{x} \in \{\mathbf{a}, \mathbf{e}\}$, normalized donor projection is

$$\pi_x(\hat{\mathbf{x}}; R, D) = \frac{(\hat{\mathbf{x}} - \mathbf{x}_R)^\top (\mathbf{x}_D - \mathbf{x}_R)}{\|\mathbf{x}_D - \mathbf{x}_R\|_2^2}. \quad (1)$$

68 The recipient is 0 and donor is 1; changes orthogonal to the donor direction receive little weight. We
 69 therefore pair projection with source retrieval: among four native candidate actions, the recomputed
 70 action must be nearest to the action paired with the installed source. Each state contributes 12
 71 off-diagonal recipient–source cells, with 25% chance under within-state label permutation. Every
 72 endpoint arm is executed after restoring the same state.

73 The matched four-source studies evaluate Cosmos 3 on 90 archival states from 30 episodes across
 74 six RoboLab tasks; FastWAM Optional-IDM on 120 states across all 40 LIBERO tasks; DreamZero
 75 on 30 deterministically selected states from distinct episodes and instructions in its released DROID
 76 training-data corpus; and LingBot-VA on 30 predetermined LIBERO-Long states across ten tasks
 77 [10, 12, 15, 17, 20–22]. No source separation or intervention outcome enters cohort selection. Cos-
 78 mos Policy retains its paired 30-state, ten-task study with physical replay [8, 16]. State is the inde-
 79 pendent unit; source directions and repeated draws are averaged within state, with model-specific
 80 frozen bootstrap intervals. Exact self/replay controls, model-specific no-op parity checks, and action-
 81 coordinate non-write audits test the implementations. Full designs appear in Appendix A.1–A.6.

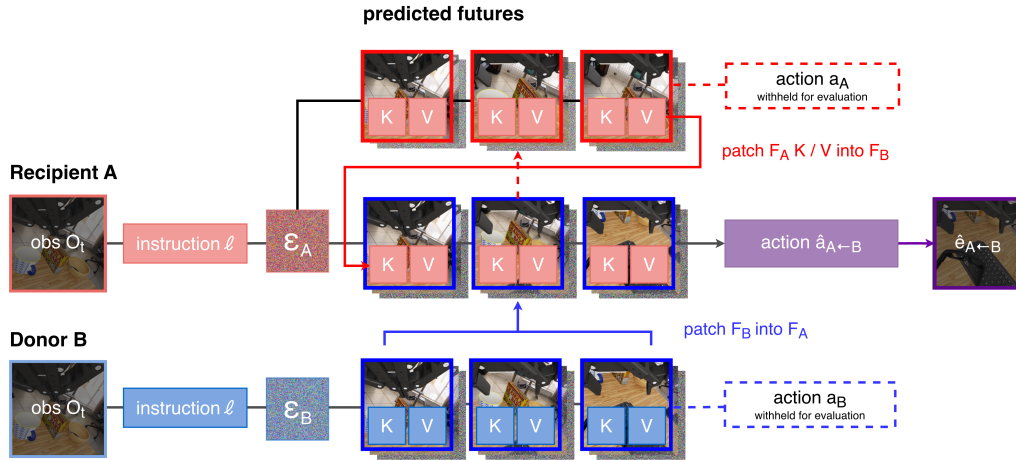


Figure 2: Stage-2 pathway intervention. We independently set visible-future identity and the source of predicted-future-position K/V read by action queries. The crossed cells test whether action follows the visible future or its K/V source.

82 2.2 Stage 2: K/V pathway intervention

83 Stage 2 tests the direct future-to-action attention route in Cosmos 3. We record predicted-future-
 84 position K/V under recipient and donor futures, then independently cross visible future F_R/F_D with
 85 K/V source $K/V_R/K/V_D$. This 2×2 design includes both suppression ($F_D, K/V_R$) and rescue
 86 ($F_R, K/V_D$), distinguishing routed source information from generic interface disruption. Cosmos
 87 Policy provides a separate prespecified key-removal test at block 27.

88 For Cosmos 3, K/V is recorded across all 36 layers and eight model forwards and replayed call
 89 by call. Only action-query attention outputs are recomputed; current-video and predicted-future
 90 query outputs remain native. Only future-position K/V are directly replaced; text, current-video,
 91 and action K/V are not overwritten. The interface was selected on excluded calibration tasks and
 92 frozen before population testing. Cache alignment, identity controls, and implementation details
 93 appear in Appendix A.3.

94 3 Results

95 3.1 Future-source identity directionally controls action

96 In the outcome-unfiltered Cosmos 3 cohort, the recomputed action identifies the installed future
 97 source in all 1,080 off-diagonal comparisons across 90 states (Table 1). Transplantation reduces
 98 distance to the source action by 0.763 ([0.733, 0.792]), with projection 0.966 ([0.956, 0.974]); the
 99 result holds in every decision phase and native-separation quartile. A separate selected-pair 22-state
 100 execution cohort links this action effect to behavior: predicted-future action steering is 0.992, and
 101 executing the recomputed actions yields endpoint steering of 1.003 ([0.950, 1.050]; 22/22 states
 102 positive).

103 The same four-source test replicates across three released systems. FastWAM’s explicit video-to-
 104 action route identifies the source in 1,439/1,440 off-diagonal cells, while its matched no-future route
 105 is exactly invariant to video seed. DreamZero’s joint within-chunk denoising identifies 360/360
 106 sources under matched per-step latent replay. LingBot-VA’s ordered future-to-cache-to-action route
 107 identifies 269/360 (0.747; [0.667, 0.819]), above the 25% permutation null; this retrieval is primarily
 108 a later-chunk effect. Thus, simultaneous video/action denoising during action generation is not
 109 required. Pathwise dose tests are graded in both new systems: the three interior recipient-to-donor
 110 mixture levels are monotone in 30/30 DreamZero and 30/30 LingBot-VA states.

Table 1: Matched four-source predicted-action results. Each state contributes 12 off-diagonal source transplants. Intervals are model-specific frozen 95% bootstrap CIs; models are not pooled.

System	States	Correct source	Distance reduction	Projection
Cosmos 3	90	1080/1080	0.763 [0.733, 0.792]	0.966 [0.956, 0.974]
FastWAM	120	1439/1440	0.683 [0.638, 0.738]	0.869 [0.833, 0.912]
DreamZero	30	360/360	0.917 [0.897, 0.933]	0.990 [0.984, 0.995]
LingBot-VA	30	269/360	0.473 [0.421, 0.524]	0.675 [0.624, 0.723]

111 Cosmos Policy supplies complementary physical evidence. At the first query of all ten LIBERO-
 112 Long tasks, paired-donor action projection is 0.499 ([0.336, 0.679]) and executed-endpoint projec-
 113 tion is 0.552 ([0.387, 0.728]), both positive in 10/10 tasks. Its near-zero effect at separately sampled
 114 later states shows strong attenuation at later sampled states.

115 3.2 Content profiles differ across architectures

116 Cosmos Policy’s early-state effect is concentrated in future visual inputs, especially the wrist cam-
 117 era. Wrist-camera replacement produces action projection 0.435 ([0.284, 0.605]), primary-camera
 118 replacement 0.101 ([0.057, 0.144]), and future-proprioception replacement 0.001 ([−0.004, 0.008]).
 119 A separate rendered-video study finds an approximately zero object-state main effect on action and
 120 endpoint, while four exploratory natural-pair comparisons favor robot-motion futures. Together
 121 these results are most consistent with camera-visible prospective robot motion.

122 Cosmos 3 requires more than either isolated pixel factor. Across ten frozen states spanning all six
 123 tasks, a complete executed donor video strongly steers actions (0.988) and endpoints (0.929), but
 124 robot-pixel effects are 0.002 and 0.010 and object-pixel effects 0.006 and approximately zero. Every
 125 isolated-factor interval lies inside the prespecified [−0.10, 0.10] equivalence region. This establishes
 126 isolated-factor insufficiency under the hybrid-mask design; the complete-video effect may depend
 127 on global context, tokenization or masking, distributed features, temporal robot–object coherence,
 128 or interactions among them (Appendix Figure 4).

129 3.3 Future-token K/V redirects Cosmos 3 action

130 The crossed 2×2 separates visible-future identity from future-token K/V source (Table 2). With re-
 131 cipient visible future, changing recipient to donor K/V increases projection by 0.895 ([0.859, 0.928]);
 132 with donor visible future, the K/V effect is 0.885 ([0.845, 0.921]). Both crossed arms follow the K/V
 133 source in all 42 state-level comparisons. Donor K/V therefore rescues donor-directed action under
 134 a recipient visible future, while recipient K/V suppresses it under a donor future. The latter retains
 135 projection 0.119 ([0.078, 0.162]), so the route is major but incomplete. In the separate 22-state ex-

Table 2: Cosmos 3 crossed future $\times$ K/V intervention across 21 states. Entries are donor-axis action projection [95% task $\rightarrow$ state hierarchical-bootstrap CI].

Visible future	Future-token K/V	Projection [95% CI]
Recipient	Recipient	-0.011 [-0.022, 0.000]
Donor	Recipient	0.119 [0.078, 0.162]
Recipient	Donor	0.884 [0.848, 0.919]
Donor	Donor	1.004 [0.988, 1.025]

136 ecution cohort, recipient-K/V restoration reduces endpoint steering by 0.851 ([0.766, 0.922]; 22/22
137 positive).

138 4 Discussion and Limitations

139 Controlled interventions turn an ambiguous co-generated output into testable claims about what
140 controls action and how that effect is transmitted. Source-specific steering recurs in Cosmos 3 and
141 DreamZero’s joint within-chunk computations and in FastWAM and LingBot-VA’s ordered video-
142 first routes. LingBot-VA therefore rules out simultaneous video/action denoising during the action
143 stage as a necessary explanation. Cosmos Policy connects steering to physical execution and shows
144 that its early-state effect is concentrated in visual future targets. The Cosmos 3 factorial then shows
145 that future-token K/V can suppress, rescue, and redirect the donor-axis action shift.

146 These results establish counterfactual control under future transplantation, not planning among al-
147 ternatives. Planning would additionally require comparing predicted consequences using a goal or
148 value and selecting among them. Our interventions instead isolate a prior question: whether chang-
149 ing a future representation redirects action toward the action associated with that source.

150 **Limitations.** The matched cross-model grids measure predicted action chunks; physical evidence
151 comes from the Cosmos cohorts and remains one-chunk rather than closed-loop success. The
152 archival Cosmos 3 grid uses lossy reconstructed inputs, DreamZero’s analysis-held-out states come
153 from its released training-data domain, and the evaluated LingBot-VA checkpoint was post-trained
154 on the tested LIBERO-Long tasks. LingBot-VA source retrieval is primarily a later-chunk effect.
155 External-model source hashes establish distinct native representations but do not by themselves
156 label semantic differences between decoded videos. Pixel masks test localized factors, and K/V
157 replacement isolates a specified interface rather than the complete circuit.

158 **Responsible use.** These methods could improve auditing by distinguishing visually plausible pre-
159 dictions from representations that actually control behavior. They could also redirect robot actions
160 or encourage false confidence on narrow state distributions. Before physical deployment, mitiga-
161 tions should include broader coverage, independent replication, physical safety constraints, human
162 oversight, and monitoring beyond the intervention metric.

163 5 Conclusion

164 Across four matched source-identification grids and complementary Cosmos execution studies, na-
165 tive future representations directionally control predicted actions under transplantation. Cosmos 3
166 future $\times$ K/V interventions identify a major internal route that carries this source-specific effect.

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A Reproducibility and Protocol Details

A.1 Experimental settings and frozen designs

The Cosmos Policy experiments use the public Cosmos-Policy-LIBERO-Predict2-2B checkpoint [16], LIBERO-Long, PyTorch 2.7 with CUDA 12.8, and two NVIDIA H100 80GB GPUs. The frozen design contains ten tasks, three state indices, eight native policy runs per state, one reciprocal donor pair per state, and three predicted-future noise draws per direction. Within each state, the pair maximizes the mean within-state rank of native normalized-action and executed-endpoint distance subject to frozen nondegeneracy thresholds; intervention outcomes never enter selection. The observation prefix, instruction, recipient action noise, predicted-future noise schedule, tokenizer settings, and denoising schedule are held fixed within an intervention comparison.

The Cosmos 3 experiments use the public 16B Cosmos3-Nano-Policy-DROID checkpoint with the released Cosmos, Cosmos Framework, and RoboLab evaluation stack. The matched four-source cohort contains 90 archival states from 30 episodes across six tasks, sampled at early, middle, and late decision points. Each state contains four native sources and all 12 ordered off-diagonal transplants; selection uses no action separation, future appearance, success, or intervention outcome. This action-only cohort reconstructs the fixed observations from lossy archival inputs. A complementary execution cohort contains 22 completed state clusters across the same six tasks; within each state, the maximum native action-L2 pair among 16 branches was frozen before intervention, subject to the registered endpoint-validity rule. The robot/object factor study contains ten states, and the crossed future $\times$ K/V cohort contains 21 existing selected-pair states.

The external-model grids follow the same four-source logic. FastWAM Optional-IDM uses repository commit `7faa7110` and 120 frozen states across all 40 LIBERO tasks. DreamZero uses commit `ab790c19` and 30 deterministically selected states from 30 distinct episodes and instruction strings in its released preprocessed DROID training-data corpus, balanced across ten predeclared instruction families without inspecting model outputs or outcomes. LingBot-VA uses commit `7c6ffa9b`, checkpoint revision `0e89d1e7`, and 30 predetermined states formed from three initial-state indices in each of ten LIBERO-Long tasks. The checkpoint was post-trained on these LIBERO-Long tasks, so this is an unselected-state test rather than held-out-task generalization. Every cohort uses four native future/action sources, all 12 ordered off-diagonal source transplants, and fixed recipient action randomness. These external grids measure generated action chunks, not simulator endpoints.

A.2 How the donor future is inserted into the recipient run

Cosmos Policy. The donor target is a clean latent encoded with the public tokenizer from the RGB and proprioceptive endpoint reached by a normal 16-action rollout. In the released latent layout, temporal indices 5, 6, and 7 are predicted future proprioception, wrist RGB, and primary RGB. At initialization and at each of the five solver evaluations ($\sigma = 80.0, 42.291, 20.972, 9.618, 4.0$), we replace only these recipient predicted-future coordinates with the donor values. The denoiser input at noise level σ is $F_D + \sigma\epsilon_R$, where ϵ_R is the recipient’s fixed predicted-future noise draw, and the returned clean- x_0 estimate is reset to F_D at the same coordinates. The action and value coordinates are never written. Thus, the donor target is reimposed at every solver evaluation rather than inserted once into a noisy state or predicted velocity. The evaluated direct-policy configuration is always conditional (its unconditioned input is `None`), so every solver call uses the wrapped x_0 function.

Cosmos 3. The prepared rectified-flow state is flattened as vision then action: vision has shape `[1, 48, 9, 33, 40]` and action `[33, 64]`. We preserve latent frame 0 and replace predicted-future video frames 1–8 with the donor target throughout denoising. A target F_D is either the final clean video latent from a normal prediction or the public-tokenizer encoding of a 33-frame video produced by executing a normal 32-action rollout. Before both model forwards at each of four UniPC calls, predicted-future vision is set to $(1-\sigma)F_D + \sigma\epsilon_R$. After guidance-3 combines the predicted velocities, we replace only the predicted-future-vision velocity by $(x_\sigma - F_D)/\sigma$, making its clean estimate F_D at the sampler boundary. The recorded sigmas are 0.999, 0.937, 0.833, 0.624. The current-video and action state and action velocity are never changed.

FastWAM Optional-IDM. The released Optional-IDM route generates a future-video representation before action denoising. For each recipient path, we keep its observation, prompt, and action RNG fixed and replace the future latent—and therefore the deterministic future-derived cache read by action denoising—with each native source. Latent and cache access are two views of the same bottleneck, not independent pathways. The matched `first_frame` mode is the no-future negative control.

DreamZero. DreamZero is autoregressive across chunks and jointly denoises video and action within a chunk. For each of 16 solver steps, we replay the donor’s naturally evolving future latent at the corresponding noise level while retaining a separately recorded recipient action-noise trace. This matches intervention strength over the denoising trajectory without writing action coordinates. Exact no-op parity compares inactive and record routes within the patched server; no separate action-array oracle from an unmodified server was retained.

LingBot-VA. The released per-chunk inference route first predicts a future-video latent, materializes it as predicted-token K/V, and then denoises the action while reading that cache. We replace the complete future-derived cache with a native source cache while fixing observation, prompt, initial latent, and recipient action noise. Once a cache override is supplied, the raw future tensor is not independently read by action generation; the intervention therefore tests the ordered future-to-cache-to-action interface.

A.3 K/V pathway implementation

The Cosmos 3 recipient and donor runs record keys and values at predicted-future video positions separately for all 36 layers and all eight model forwards: four denoising calls, each with conditional

and unconditional passes. The crossed factorial independently installs recipient or donor visible future and recipient or donor K/V source. Cache entries are replayed call by call, and only action-query rows are recomputed. Query outputs at current-video and predicted-future-video positions remain native. Only future-position K/V are directly replaced; text, current-video, and action K/V are not overwritten. A calibration scan on excluded tasks selected layer 16 for single-layer localization, whereas the population estimate replaces all 36 direct K/V interfaces from predicted-future positions to action queries.

A.4 State restoration, execution, and missingness

Each endpoint comparison begins from an identical saved simulator state. Evaluation reconstructs the environment, restores that state, replays the observation prefix, verifies the restored observation, and then executes one recomputed action chunk. Recipient and donor endpoints are produced by the same restoration and execution procedure. This design makes the endpoint projection a comparison between reachable continuations of one physical state rather than between unrelated trajectories.

Two frozen Cosmos 3 population units were unavailable because their registered source branches ended after 21 and 29 actions, before producing the required 32-action, 33-frame intervention source. Neither unit was replaced, and no intervention outcome entered the decision to retain or omit a state. The completed population therefore exceeds the prespecified minimum of 20 states across at least five tasks. The factor study likewise meets its minimum of ten states across at least five tasks.

A.5 Intervention validity and controls

For Cosmos Policy, the self condition passes the recipient target through the complete intervention machinery. For Cosmos 3, the primary self reference is an exact native recomputation. The mechanics-matched clean self-clamp has mean donor projection -0.006 ; donor-minus-self-clamp steering remains 0.999 and is positive in 22/22 states. Identity token-layout capture and K/V record/replay are bitwise exact relative to the same clamped-self trajectory. Sentinels at the sampler boundary confirm that replacing recipient predicted-future coordinates with donor values does not write directly into action coordinates. The pathway experiment preserves token count and order and directly overwrites only predicted-future-position K/V; current-video, language, and action K/V positions are not directly overwritten.

The geometry-matched Gaussian control has the same norm and recipient distance as the donor displacement but no native source identity. Natural controls are valid unpaired alternative future targets selected without intervention outcomes, and shuffled controls use another target from the same task; the Cosmos 3 natural target is a model-predicted future. Exact self and replay controls test implementation drift, while native-source retrieval distinguishes directional identity from displacement alone. Gaussian interventions additionally expose off-manifold sensitivity and are not assumed to preserve the recipient. Executed-video conditions provide a realizability check by encoding frames generated when a native donor action is physically replayed in the simulator.

A.6 Stage-1 statistical analysis

The independent unit is a state, not an individual source direction or noise draw. In four-source grids, all 12 off-diagonal directions are averaged within state. Cosmos 3 intervals use task→episode/state hierarchical resampling, and FastWAM uses suite→task→state resampling. DreamZero, LingBot-VA, and Cosmos Policy resample states directly; family, task, and leave-one-group summaries are separate sensitivity analyses. Cosmos Policy reciprocal directions and repeated draws are averaged within state. Reported 95% intervals use each model’s frozen percentile-bootstrap scheme [4]; source-retrieval tests use within-state label permutation. Models are analyzed separately rather than pooled.

We prespecified 0.10 projection units—10% of the native recipient-to-donor displacement—as the smallest effect of interest for the Cosmos Policy confirmatory family and the Cosmos 3 population and factor families [9]. An interval contained within $[-0.10, 0.10]$ supports practical equivalence to zero at that scale. Prespecified primary sign tests use state-level directions and Holm correction within each family.

379 A.7 Stage-2 statistical analysis

380 Stage 2 reuses the Stage-1 independent-unit and bootstrap conventions. For Cosmos 3, the effect of
 381 K/V source is formed within each of 21 states separately under recipient and donor visible futures.
 382 Effects are averaged equally across tasks and assigned task→state hierarchical-bootstrap intervals.
 383 The crossed source-identification comparison asks whether F_R , K/V_D moves toward the donor and
 384 F_D , K/V_R toward the recipient. The separate Cosmos Policy key-removal study uses ten first-query
 385 states; its endpoint intervals are reported in Appendix B.7.

386 B Additional Experiments and Diagnostics

387 B.1 Stage-1 directional steering

388 Figure 3 displays the Stage-1 estimates at the state or task level. Panel a shows donor-minus-self
 389 steering, while panel b shows the paired donor’s advantage over each control. The unit-level markers
 390 expose variation suppressed by the aggregate estimates reported in the main text.

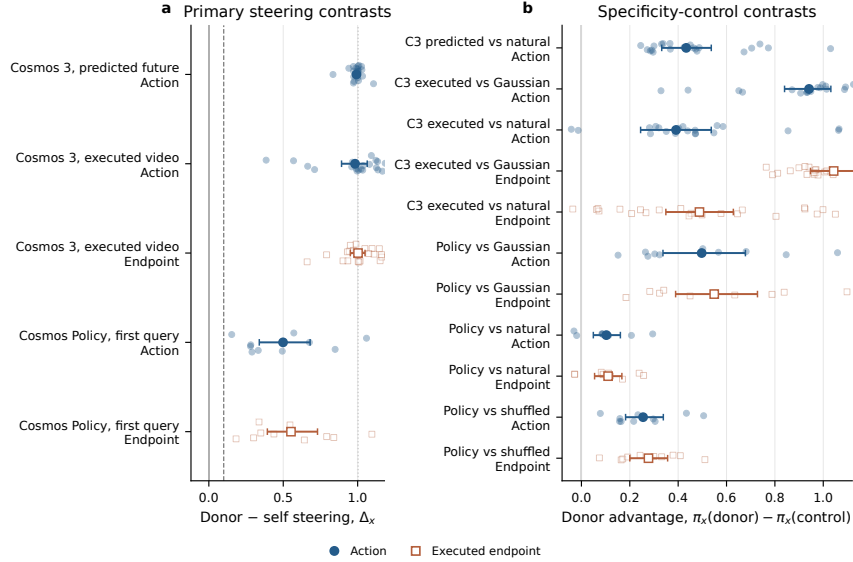


Figure 3: Stage-1 directional steering (a) and donor-minus-control contrasts (b). Small markers are states for Cosmos 3 and tasks for Cosmos Policy; large markers and whiskers are means and 95% bootstrap intervals. Blue denotes actions and orange denotes endpoints.

391 B.2 Matched four-source action grids

392 The Cosmos 3 outcome-unfiltered cohort achieves 1,440/1,440 correct source identifications across
 393 all cells and 1,080/1,080 off-diagonal cells. Off-diagonal normalized orthogonal residual is 0.227
 394 ([0.200, 0.255]). Retrieval is perfect in every early/middle/late phase and every prespecified native-
 395 separation quartile. Exact native, self, donor, and no-op replays pass, and sentinels verify zero writes
 396 to action coordinates. The cohort adds action evidence from archival inputs; endpoint evidence
 397 remains attached to the independent 22-state execution cohort.

398 FastWAM achieves 1,919/1,920 all-cell and 1,439/1,440 off-diagonal identifications, with cosine
 399 alignment 0.943 ([0.927, 0.962]). The matched `first_frame` route is bitwise invariant to donor-
 400 video seed. DreamZero achieves 480/480 all-cell and 360/360 off-diagonal identifications; co-
 401 sine alignment is 0.995 ([0.991, 0.997]) and orthogonal residual 0.080 ([0.065, 0.099]). Its first
 402 low-level action retains 348/360 correct off-diagonal identifications. At future-mixture strengths
 403 $\alpha = 0.25, 0.50, 0.75$, mean DreamZero projection is 0.046, 0.465, 0.939; all 30 state-level interior
 404 profiles are monotone, with slope 1.786 ([1.707, 1.853]); one-sided sign-flip $p = 0.00001$.

LingBot-VA achieves 389/480 all-cell and 269/360 off-diagonal identifications, with cosine alignment 0.814 ([0.777, 0.847]) and orthogonal residual 0.388 ([0.359, 0.418]). Its full action chunk carries the effect: retrieval is 98/360 for the first post-conditioning frame group and 48/360 for the literal first low-level action. Exact routing controls give projection 0 for donor raw future with recipient cache and 0.675 ([0.624, 0.723]) for recipient raw future with donor cache, confirming that the action follows installed cache identity. At $\alpha = 0.25, 0.50, 0.75$, mean projection is 0.221, 0.487, 0.758; all 30 state-level interior profiles are nondecreasing, with slope 1.073 ([1.039, 1.109]; $p = 0.00001$). The interpolation tests graded pathwise sensitivity along fixed latent segments; they do not assign semantic meaning to α .

B.3 Cosmos 3 intervention timing and strength

A separate outcome-unfiltered 30-state timing audit shows that the effect does not require clamping at every solver call. Clamping only the final of four calls gives complete source retrieval of 1.000 and distance reduction 0.761 ([0.715, 0.801]), compared with 0.760 ([0.714, 0.801]) when all calls are clamped. In a five-level future-strength audit on 30 states, mean projection at $\alpha = 0, 0.25, 0.50, 0.75, 1$ is $-0.002, 0.056, 0.303, 0.877, 0.969$. The prespecified distance-reduction slope is 0.899 ([0.852, 0.936]), and every adjacent population contrast is positive. Thus donor-directed action changes with both intervention timing and strength rather than appearing only under the sustained full-strength condition.

B.4 Cosmos 3 robot/object factorization diagnostic

Figure 4 shows the full 2×2 hybrid intervention summarized in Section 3.2. Panels a and b retain the four pixel-source cells and state-level estimates; panel c places the two factorial main effects beside the full executed-donor-video benchmark.

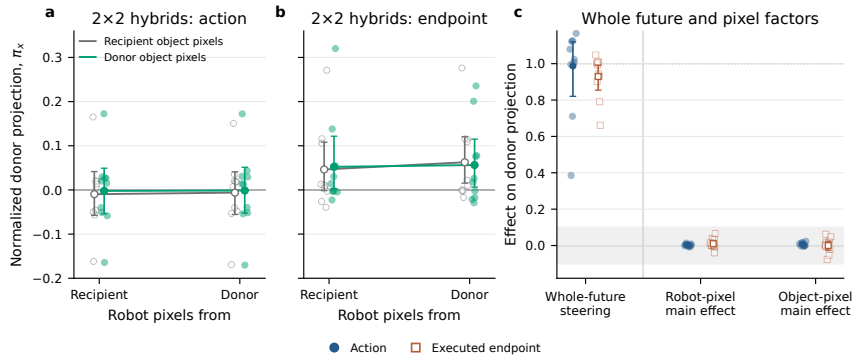


Figure 4: Cosmos 3 robot/object pixel factorization. Panels (a)–(b) show the four hybrid-video cells; panel (c) compares robot/object main effects with the full executed-donor-video benchmark. Small markers are states; large markers and whiskers are means and 95% bootstrap intervals.

B.5 Condition-level Cosmos 3 projections

Figure 5 shows the condition estimates before they are converted into donor-minus-control contrasts. Recomputation with the recipient’s own predicted future stays at zero, naturally predicted alternatives produce partial donor-directed movement, and paired donors cluster near one. For executed videos, the matched Gaussian remains near zero while both action and endpoint effects preserve the same ordering. The state-level display also makes clear that the endpoint conclusion is not produced by a single task or a single extreme unit.

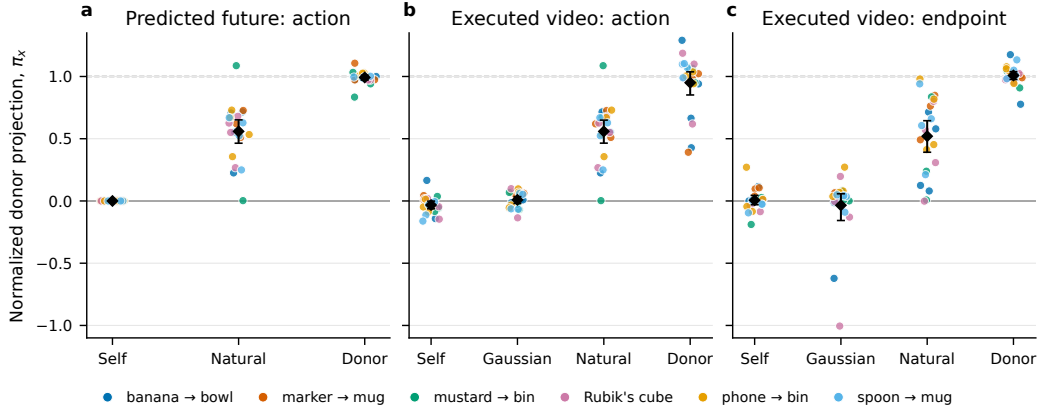


Figure 5: Cosmos 3 condition-level projections across 22 restored states for predicted actions, executed actions, and endpoints. Colors identify tasks; black diamonds and whiskers are means and 95% state-bootstrap intervals.

434 B.6 Displacement-magnitude diagnostic

435 Directional projection could be misleading if a large normalized estimate arose from an extremely
 436 small native denominator or if inserting the donor's predicted future caused only a negligible displacement.
 437 Figure 6 therefore compares unnormalized recipient-to-donor separation with the displacement
 438 caused by replacing the recipient's predicted future with the donor's. Predicted-action and
 439 executed-endpoint displacements lie close to the identity line across the observed range; executed-
 440 action displacement is more variable but remains on the scale of the native difference. This diagnostic
 441 supports the interpretation that the intervention recreates a substantial behavioral displacement,
 442 not merely a tiny change that happens to align with the donor axis.

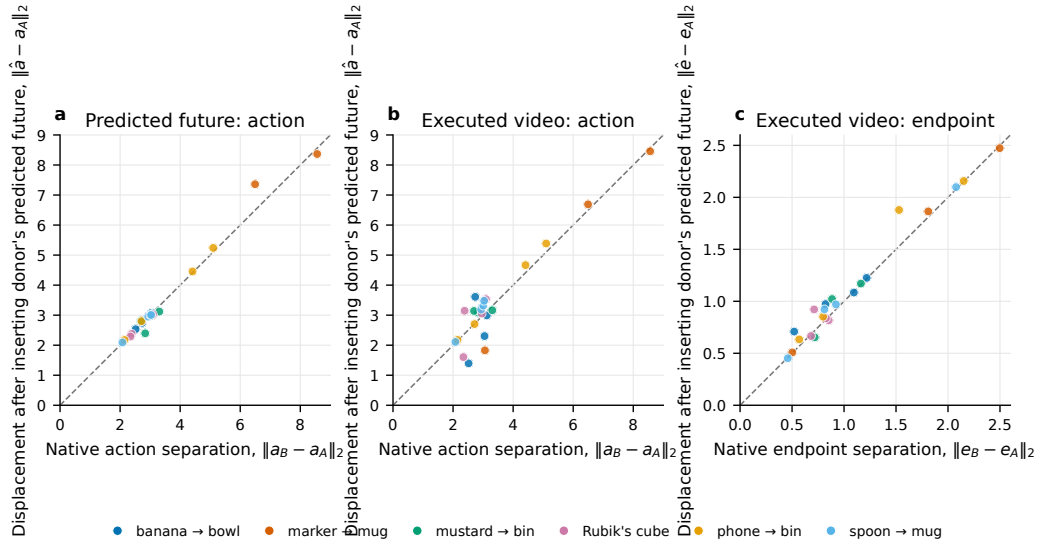


Figure 6: Native recipient-to-donor separation versus predicted-future-insertion displacement for Cosmos 3 predicted actions, executed actions, and endpoints. Dashed lines indicate equality; colors identify tasks.

443 B.7 Cosmos Policy timing and cross-domain checks

444 Across all 30 registered Cosmos Policy states, mean donor-minus-self steering is 0.186
 445 $([0.093, 0.290])$ for actions and 0.194 $([0.091, 0.305])$ for endpoints. The first-query, after-three-
 446 chunk, and after-six-chunk action estimates are 0.499, 0.031, and 0.027; the corresponding endpoint
 447 estimates are 0.552, 0.004, and 0.025. Because the three indices represent different physical states
 448 that may be nested within task trajectories, the first-query analysis is the cleaner confirmatory result
 449 and the differences should not be read as causal temporal decay.

450 The six-state RoboCasa replication is exploratory and is not pooled with LIBERO [14]. Donor-
 451 minus-self action steering is 0.0144 $([0.0034, 0.0276])$, while its donor-minus-Gaussian contrast is
 452 0.0025 with an interval crossing zero. Endpoint steering is 0.0287 $([0.0101, 0.0492])$, positive in 6/6
 453 states; donor-minus-Gaussian endpoint steering is 0.0187 $([0.0063, 0.0329])$, also positive in 6/6.

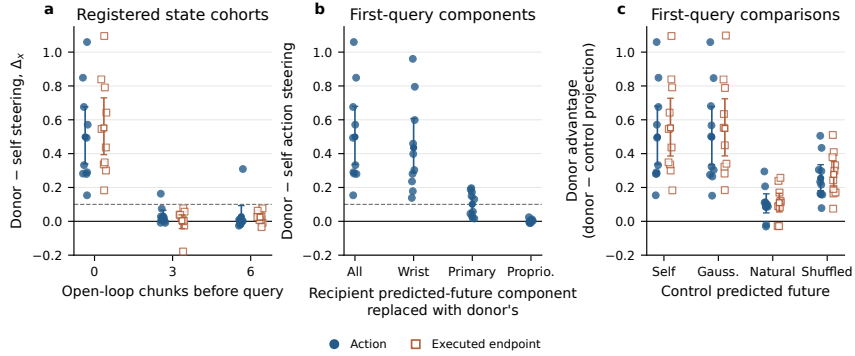


Figure 7: Cosmos Policy scope checks: (a) state cohorts, (b) future-target components, and (c) donor-minus-control contrasts. Small markers are tasks; large markers and whiskers are means and 95% task-bootstrap intervals.

454 B.8 Within-task relationships in Cosmos Policy

455 Figure 8 provides two task-level diagnostics at the first query. Action and executed-endpoint steering
 456 generally rise together, supporting the use of simulator execution as a behavioral continuation of the
 457 action result. Steering from the wrist-camera component also tracks steering from the full future
 458 target across tasks, consistent with the aggregate modality decomposition while showing that the
 459 wrist view does not reproduce the entire effect uniformly.

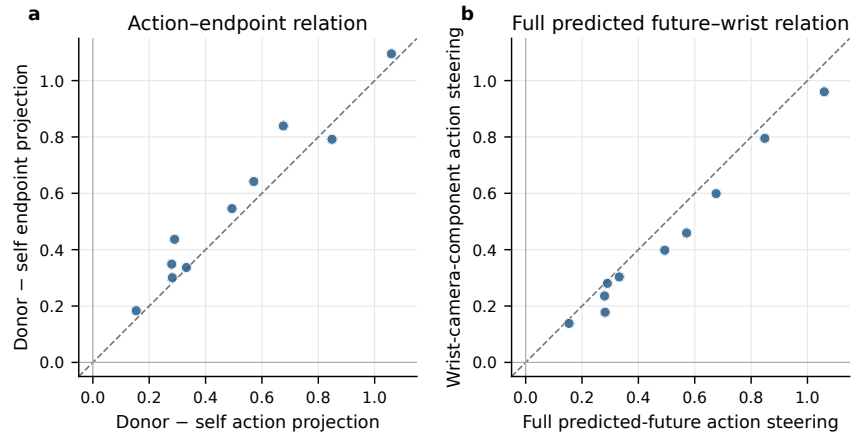


Figure 8: Cosmos Policy task-level relationships at the first query: (a) action versus endpoint steering and (b) full-target versus wrist-camera-component steering. Points are tasks; dashed lines indicate equality.

460 B.9 Additional content and pathway checks

461 The Cosmos Policy rendered 2×2 study supplies a manipulation check for the object-content null.
 462 Decoded future targets identify the intended hybrid cell with 92.5% accuracy in the primary camera
 463 and 98.75% in the wrist camera, while executed endpoints identify the four cells at the 25% chance
 464 rate. Thus, the future representation visibly contains the imposed distinction even though the pre-
 465 registered object main effect on action and goal-state endpoint is approximately zero. Four states
 466 met both natural-pair criteria; in these comparisons, robot-motion donors steer more strongly than
 467 object-motion donors.

468 The block-27 Cosmos Policy key-removal analysis includes two additional checks beyond the action-
 469 space dose curve reported in the main text. Removing all keys at predicted-future positions changes
 470 the executed endpoint by an absolute mean projection of 0.281 ([0.164, 0.430]; 10/10 first-query
 471 states) and exceeds equal-count random-key removal by 0.169 ([0.058, 0.322]; 9/10). Its difference
 472 from equal-count current-key removal is -0.035 with an interval crossing zero. The endpoint anal-
 473 ysis therefore supports an active route from the predicted future at the first query while reinforcing
 474 the conclusion that this route is not uniquely dominant over current-state information.

475 For Cosmos 3, the crossed future $\times$ K/V study gives equal-task cell projections -0.011 for recipient
 476 future/recipient K/V, 0.119 for donor future/recipient K/V, 0.884 for recipient future/donor K/V, and
 477 1.004 for donor future/donor K/V. The corresponding K/V-source effects are 0.895 ([0.859, 0.928])
 478 and 0.885 ([0.845, 0.921]). This conditional rescue and redirection strengthens the pathway inter-
 479 pretation beyond suppression alone. In the separate execution cohort, task-balanced pathway-loss
 480 estimates are 0.877 for predicted actions, 0.812 for executed-video actions, and 0.853 for executed
 481 endpoints. The positive residual under recipient K/V motivates tests of parallel attention heads,
 482 non-attention routes, and interactions between patched and unpatched components.

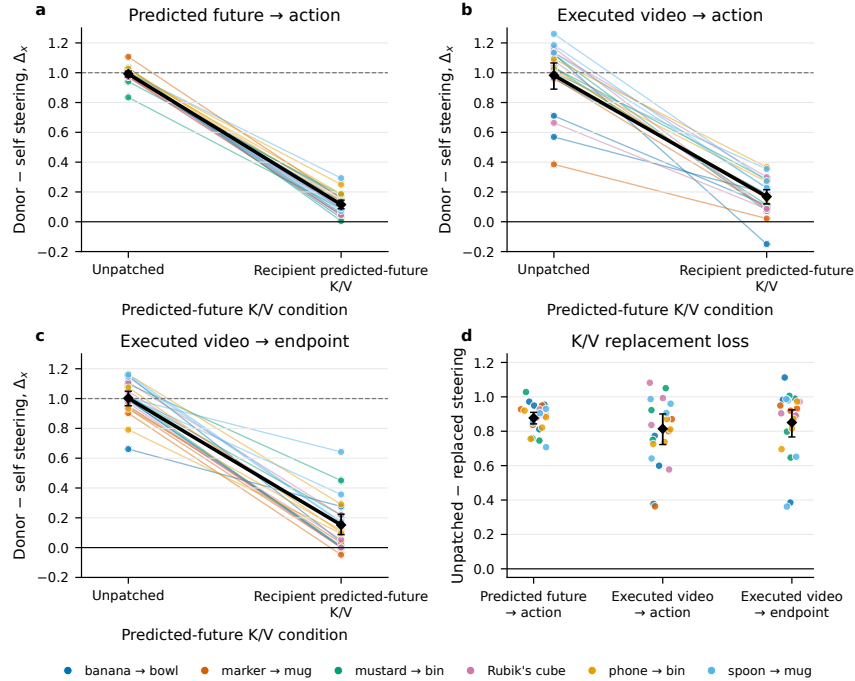


Figure 9: Cosmos 3 donor steering before and after K/V replacement at predicted-future token positions. Lines connect the same state; black diamonds and whiskers are means and 95% state-bootstrap intervals. Panel (d) shows replacement loss.